

Software Requirements Specification

Short Shelf Life

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1 Introduction

1.1 Purpose

This document describes the functional and non-functional requirements for the Short Shelf Life Application.

1.2 Project Description

The system allows users to create and manage books in their bookshelves, leave reviews, and view bookshelves of other users.

2 Functional Requirements

2.1 User Registration

Description: The system shall allow users to create an account using an email address and password only through valid email providers. Default profile state is public.

Priority: High

Input:

- Shelf Name
- Email address
- Password (salted hash) (length 30 characters, must contain lower and upper case letters and a special character)
- Re-enter Password

Output:

- Successful account creation message.
- Redirection to main bookshelf page.
- If the shelf has been populated, the books will be saved to the database instead of local storage.

Validation Rules:

- The email address must follow standard email format.
- Temporary or invalid email domains shall not be accepted.
- The system shall verify email uniqueness.

2.2 User Login

Description: The system shall allow users to login with their verified email and passwords.

Priority: High

Input:

- Email address
- Password

Output:

- Successful login message
- Access to personal bookshelf
- Authentication tokens shall be securely stored in HTTP-only cookies.
- Authentication tokens will expire after a defined amount of time (24hr).

2.3 User Password Reset

Description: The system shall allow users to reset their passwords through an OTP sent to the user's emails.

Priority: High

Input:

- New password (same constraints apply)
- Re-enter password.

Output:

- Successful password updated message.
- Redirection to login page.

2.4 User Profile View

Description: The system shall allow users to view their profile details.

Priority: High

2.5 User Profile Edit

Description: The system shall allow users to edit their email, password, and shelf names (all optional fields), privacy status (Enum public, private).

Priority: High

Input:

- Email address (optional)
- Password (optional)
- Re-enter password (if Password changing)
- Shelf name
- Private/public toggle

Output:

- Successful profile edit message

2.6 User Delete Profile

Description: The system shall allow users to delete their profiles. All data will subsequently be erased from their local storage and database both.

Priority: High

2.7 Add Books

Description: The system shall allow users to add a new book to their shelf. The default storage is the user's local storage.

Priority: High

Input:

- Book Name (0-30 characters, mandatory)
- Author Name (0-30 characters, mandatory)
- Review (0- 1300 characters, optional, default value empty)
- Color of book spine (from a color wheel, optional, default in place).
- Design of book spine (from an enum of designs, optional, default in place).

Output:

- Successfully added book message.

- Book will now be displayed on the shelf.
- Responsive design for long titles on the book spine.
- For longer titles, show the title with an ellipsis.

2.8 View Books

Description: The system shall allow users to view the books they have added. If no review is added, a closed version will be shown. If a review is added, an open version will be shown. If no books are present, appropriate message must be shown and some animated effect (fly buzzing around etc).

Users can filter books based on whether they have a review or not. Users can search for specific books based on their title or author name as well. **Priority:** High

2.9 Edit Books

Description: The system shall allow users to edit the title, author, and review of the book. The edit button will be placed on the Viewed Book, open and closed, depending on review status. Form constraints will still hold as outlined in the Add Book section.

Priority: High

Input:

- Edited Title (optional)
- Edited Author (optional)
- Edited Review (optional)

2.10 Delete Book

Description: The system shall allow users to delete the book they have entered. Deletion would remove the book from local storage, database (wherever it is saved) and the shelf.

Priority: High

Output:

- Successful book deletion message

2.11 Browse Other Shelves

Description: The system shall allow users to view other shelves. Browsing will be done through a page where all the shelves and the name of shelves are displayed. Back button will take the user back to the browsing page then the user's own shelf. Users can interact

with other shelves and read their reviews as well. If no books are present, appropriate message must be shown and some animated effect (fly buzzing around etc).

Users can filter books based on whether they have a review or not. Users can search for specific books based on their title or author name as well.

Users can only view shelves listed as public. **Priority:** Medium

2.12 Interactive Effects

Description: The system shall allow users to interact with the main page in capacity outlined below.

Priority: Low

1. A toggle switch to turn light off and on.
2. Hovering over a book would highlight it.
3. Hovering over a book would show its full title.

2.13 Animation

Description: The system shall implement animation effects in some capacity, outlined below.

Priority: Low

1. Moving leaves.
2. Moving clouds during night.
3. Moving clouds during day.
4. Birds flying during day.

2.14 Background Music

Description: The system shall play a soft music in the background, with the choice to turn it off.

Priority: Low

Priority: Input

1. A toggle switch to turn music off and on.

2.15 System Time Background

Description: The system shall change the background according to the time in PKT. From 7pm to 6am the background would reflect night time. From 6.01 am to 6.59pm the background would reflect day time.

Priority: Low

3 Non Function Requirements

The following constraints hold for the entire system where applicable:

- The system shall protect against SQL injection attacks using parameterized queries.
- The system shall sanitize user-generated content to reduce XSS vulnerabilities.
- The system would only support english language script, no other language.

4 Screens

4.1 User Account Creation

Would require Name, Email, Password.

4.2 Successful Account Creation

After 3.1, depending on outcome.

4.3 Error During Account Creation

After 3.1, depending on outcome.

4.4 User Login Page

Would require Email, Password.

4.5 Error During Account Login

After 3.4, depending on outcome.

4.6 Book Shelf Page

1. Animation effects.
2. Interactive effects.
3. Button for music toggle.
4. Button for account creation.

5. Button for About.
6. Button to Add New Book.
7. Clicking on a Book would open the book. If no review, closed view. If review, open view.
8. View Other Shelves button.
9. Name of shelf visibly displayed.
10. Logout button
11. Edit Shelf Name and Password button.
12. Search for book based on title or author.
13. Filter books based on review presence.

4.7 Add Book

Would require Title, Author, and ask for Review, Design.

4.8 View Book, Closed View

Would have the title and the author. Would also have an Update and Delete Button.

4.9 View Book, Open View

Would have the title, author and the review. Would also have an Update and Delete Button.

4.10 Successfully Added Book

After 3.7, depending on the outcome.

4.11 Error While Adding Book

After 3.7, depending on the outcome.

4.12 Edit Book

Would allow users to edit the title, author and review. Would also have a back button.

4.13 Successfully Updated Book

After 3.12, depending on the outcome.

4.14 Error While Updating Book

After 3.12, depending on the outcome.

4.15 Back Button Confirmation Message

If the user presses on the back button during editing.

4.16 Delete Book

Would allow users to delete the book. Confirmation will be asked first.

4.17 Successfully Deleted Book

After 3.16, depending on the outcome.

4.18 Error While Deleting Book

After 3.16, depending on the outcome.

4.19 Deletion Confirmation Message

Deletion will proceed only if the user presses Yes.

4.20 Other Shelves Page

A consolidated page with other shelves modeled as books. Each shelf would show the user name. Clicking on the shelf would allow you to see the other shelf. Other shelf would have a back button to return to the browse page. Browse Page would allow you to return to your shelf. Search to search for shelves.

4.21 Edit Profile Page

Users can edit shelf name and password. Will also have a back button to cancel edits.

4.22 Confirmation of Changes

Edits will only be saved if user clicks yes.

4.23 Back butto confirmation

Edits will be discarded if user clicks yes.

4.24 Main Landing Page

Will feature three options: Login, Register, Proceed as Guest.

4.25 User Profile View

Users can view profile details

4.26 Delete profile

Screens 4.17, 4.18, and 4.19 applicable here.